

Question Paper

Exam Date & Time: 25-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

SUBJECT: ECE 2125/ECE_2125 ELECTROMAGNETIC WAVES
(SPL: ELECTRONICS AND COMMUNICATION ENGINEERING)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Given $W = x^2y^2 + xyz$. Compute ∇W and the directional derivative $\frac{dW}{dl}$ in the direction $3\hat{x} + 4\hat{y} + 12\hat{z}$ at $(2, -1, 0)$. (4)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1B) Transform $\vec{F} = 10\hat{x} - 8\hat{y} + 6\hat{z}$ at point $P(10, -8, 6)$ to cylindrical coordinates. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1C) If 3 identical charges each of 'a' are assembled along a line such that the distance between successive charges is '1m'. In this case, energy stored in the system is 'W1'. Now if the distance between the successive charges is changed to 0.5m. In this case energy stored in the system is 'W2'. Establish the relation between W1 and W2. (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2A) A circular disk of radius 'a' is uniformly charged with $\rho_s \text{ C/m}^2$. The disk lies on the $z=0$ plane with its axis along z-axis. Show that at point $(0, 0, h)$, $E = \frac{\rho_s}{2\epsilon_0} \left\{ 1 - \frac{h}{(h^2 + a^2)^{1/2}} \right\} \hat{z}$ (4)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2B) Consider a sphere of radius 'a' with a uniform charge $\rho \text{ C/m}^3$. Determine $\vec{D}$ for (i) $r \geq a$ and (ii) $r \leq a$. (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2C) -9nC of charge is placed at the origin. Find $\vec{D}$ at $(3, 4, 0)$ m due to that point charge. (1 unit=1meter). (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3A) Calculate $\nabla \times [\nabla(\nabla \cdot \mathbf{H})]$ if $\mathbf{H} = 2x^2yz \mathbf{a}_x - 20y \mathbf{a}_y + (x^2 - z^2) \mathbf{a}_z$. (4)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 3B) Derive an expression for magnetic field intensity at a point due to a finite conductor carrying a current along z axis. (3)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 3C) Evaluate both sides of Stoke's theorem for the field $\mathbf{H} = 6xy \mathbf{a}_x - 3y^2 \mathbf{a}_y \text{ A/m}$ and the rectangular path around the region $2 \leq x \leq 5$, $-1 \leq y \leq 1$, $z = 0$. Let the positive direction of $d\mathbf{s}$ be $\mathbf{a}_z$. (3)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 4A) Show that the net electromagnetic power flowing out of V through its surface equals the time rate of decrease of stored electromagnetic energy in V minus the Ohmic power dissipated in V , where V is the Volume. (4)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4B) A wave propagating in lossless dielectric has the components $E = 500 \cos(10^7 t - \beta z) \mathbf{a}_x$ V/m and $H = 1.1 \cos(10^7 t - \beta z) \mathbf{a}_y$. If the wave is travelling at $v = 0.5c$, find (3)
- (a) μ_r
(b) ϵ_r
(c) β

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4C) Show that the wave impedance of a wave in free space is equal to 120π or 377 ohm. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5A) A $100 + j150\Omega$ load is connected to a 75Ω lossless line. Using Smith Chart, find (a) reflection coefficient, (b) VSWR (c) Z_{in} at 0.4λ from the load (d) The locations of V_{max} and V_{min} with respect to the load if the line is 0.6λ long (e) Z_{in} at the generator. (4)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- 5B) A distortionless line operating at 120 MHz has $R = 20\Omega/m$, $L = 0.3\mu H/m$, and $C = 63\text{ pF/m}$. Determine (a) the propagation constant, (b) characteristic impedance Z_0 and (c) the phase velocity. (3)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- 5C) Using necessary equations and diagram, explain how single stub matching is achieved in a transmission line. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 ECE_2125:ELECTROMAGNETIC WAVES

#	Course Outcome	Description	Marks
1	1	CO 1	7
2	2	CO 2	13
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 ECE_2125:ELECTROMAGNETIC WAVES

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	1	3(6%)
3	Application	11	37(74%)
4	Analysis	3	10(20%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)